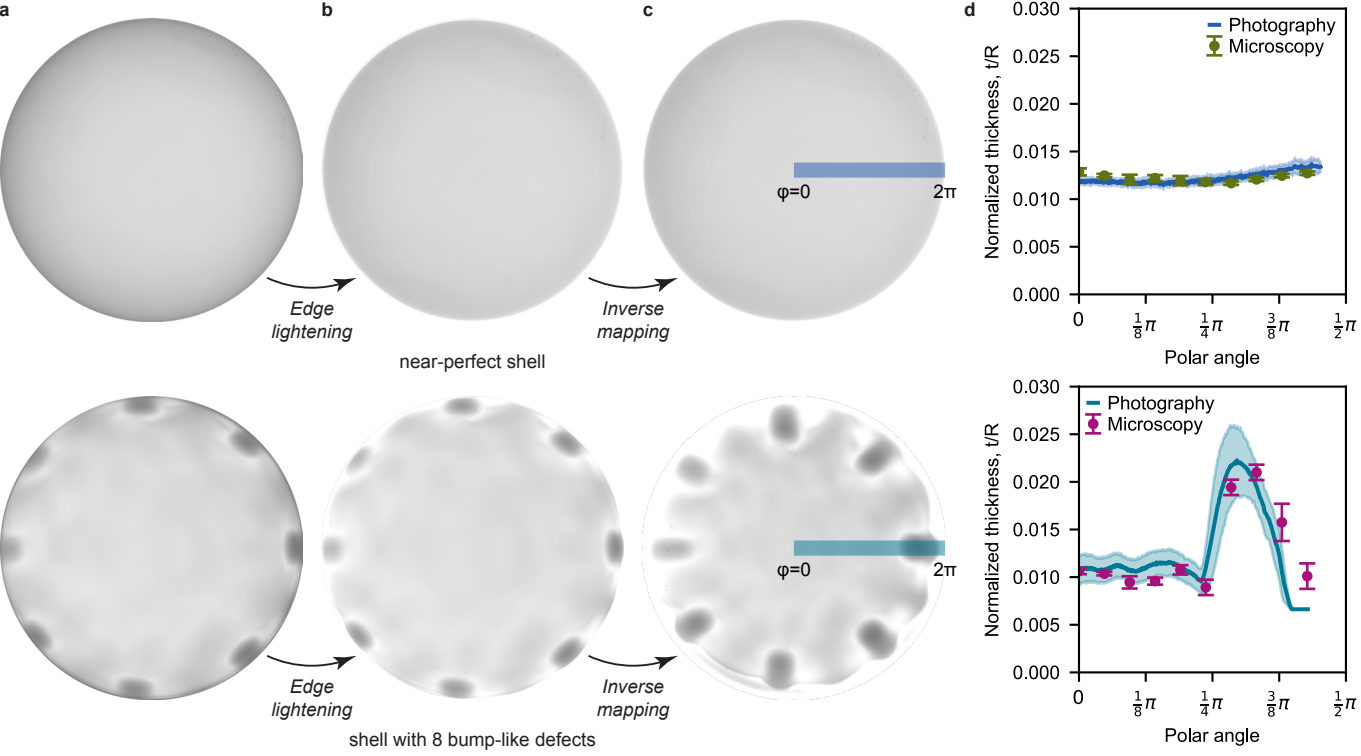
a
b
c
d
Edge lightening
Inverse mapping
φ=0
2π
near-perfect shell
Edge lightening
Inverse mapping
φ=0
2π
shell with 8 bump-like defects
Photography
Microscopy
Normalized thickness, t/R
0.030
0.025
0.020
0.015
0.010
0.005
0.000
0
1/8 π
1/4 π
3/8 π
1/2 π
Polar angle
Photography
Microscopy
Normalized thickness, t/R
0.030
0.025
0.020
0.015
0.010
0.005
0.000
0
1/8 π
1/4 π
3/8 π
1/2 π
Polar angle